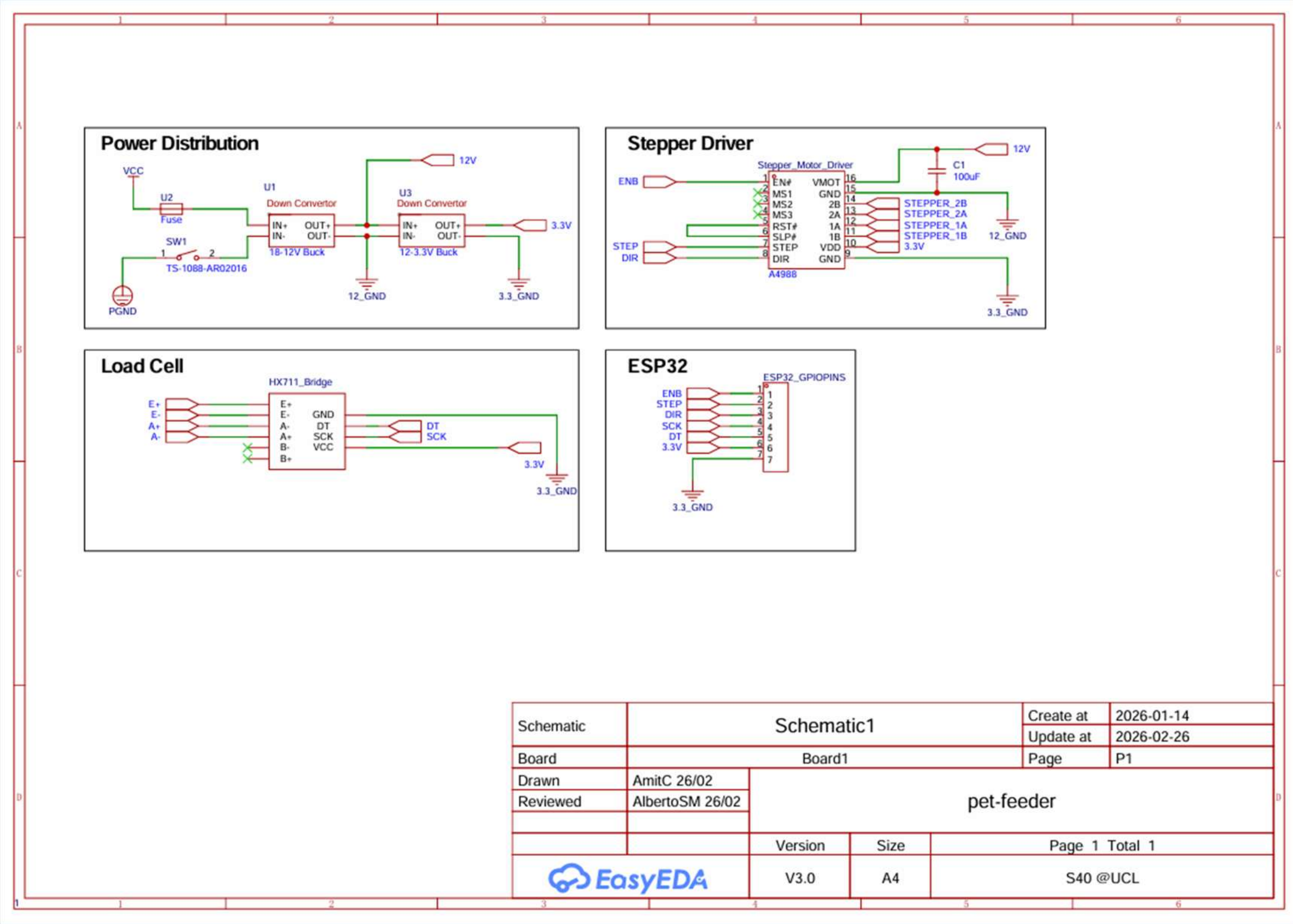
Open-Source Automatic Pet Feeder

S40

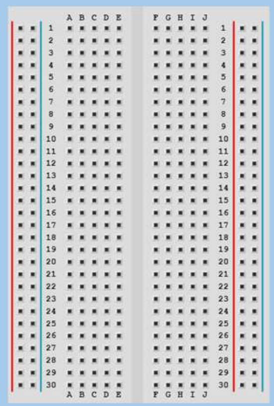
Assembly Manual

1. Breadboard Assembly



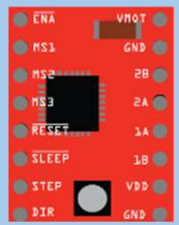
Electrical Components

Not to scale



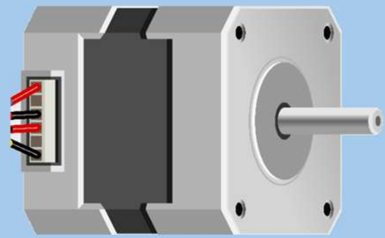
2x

Mini Breadboard



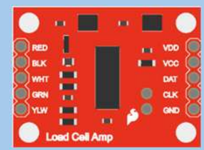
1x

A4988 Stepper Driver



1x

Nema 17 Stepper Motor

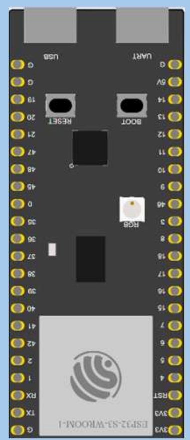


1x

HX711 Wheatstone Bridge and Op-Amp

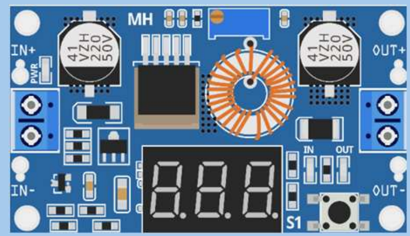
21x

Jumper Cables (varying length)



1x

ESP32-S3 Microcontroller



2x

Buckboost

1x

$100\mu F$ Decoupling Capacitor

1x

5A Fuse

1x

Load Cell

Breadboard Wiring

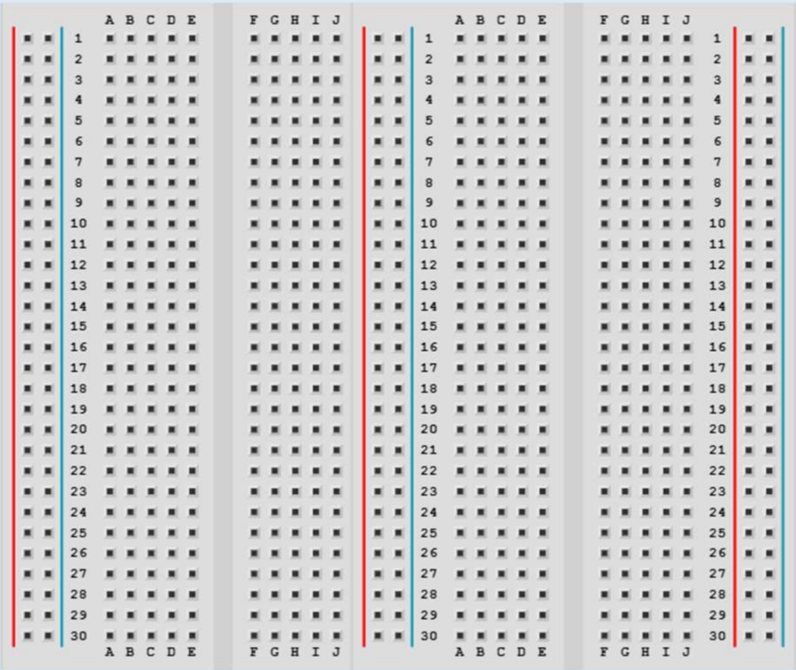
Breadboard Assembly

2x

15x

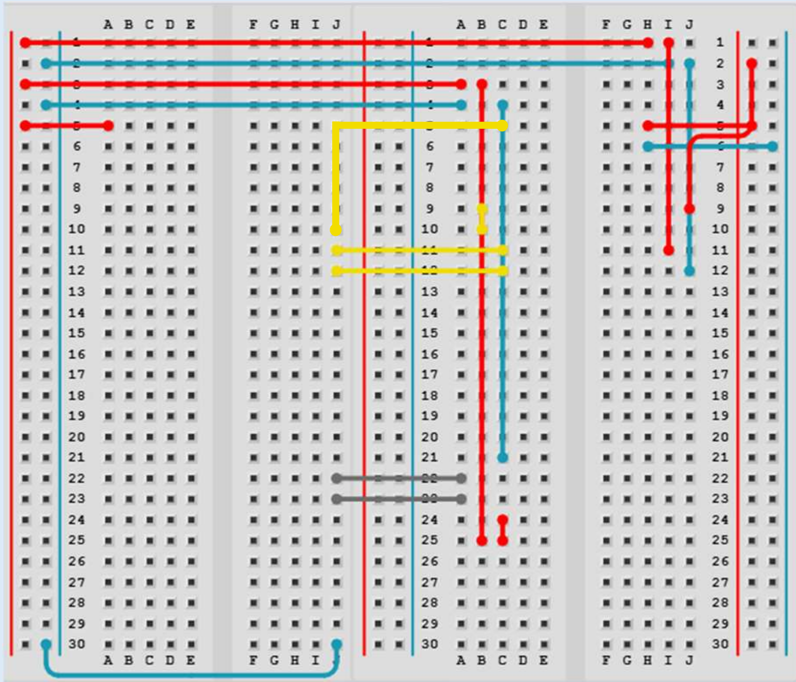
1\.

Connect the two mini breadboards together as shown (no middle rails necessary)



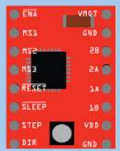
2\.

Place all jumper wires as shown



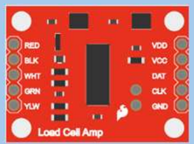
Component Placement

Breadboard Assembly



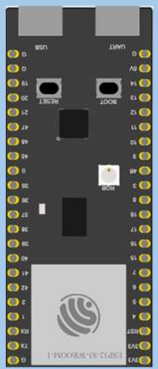
1x

A4988



1x

HX711



1x

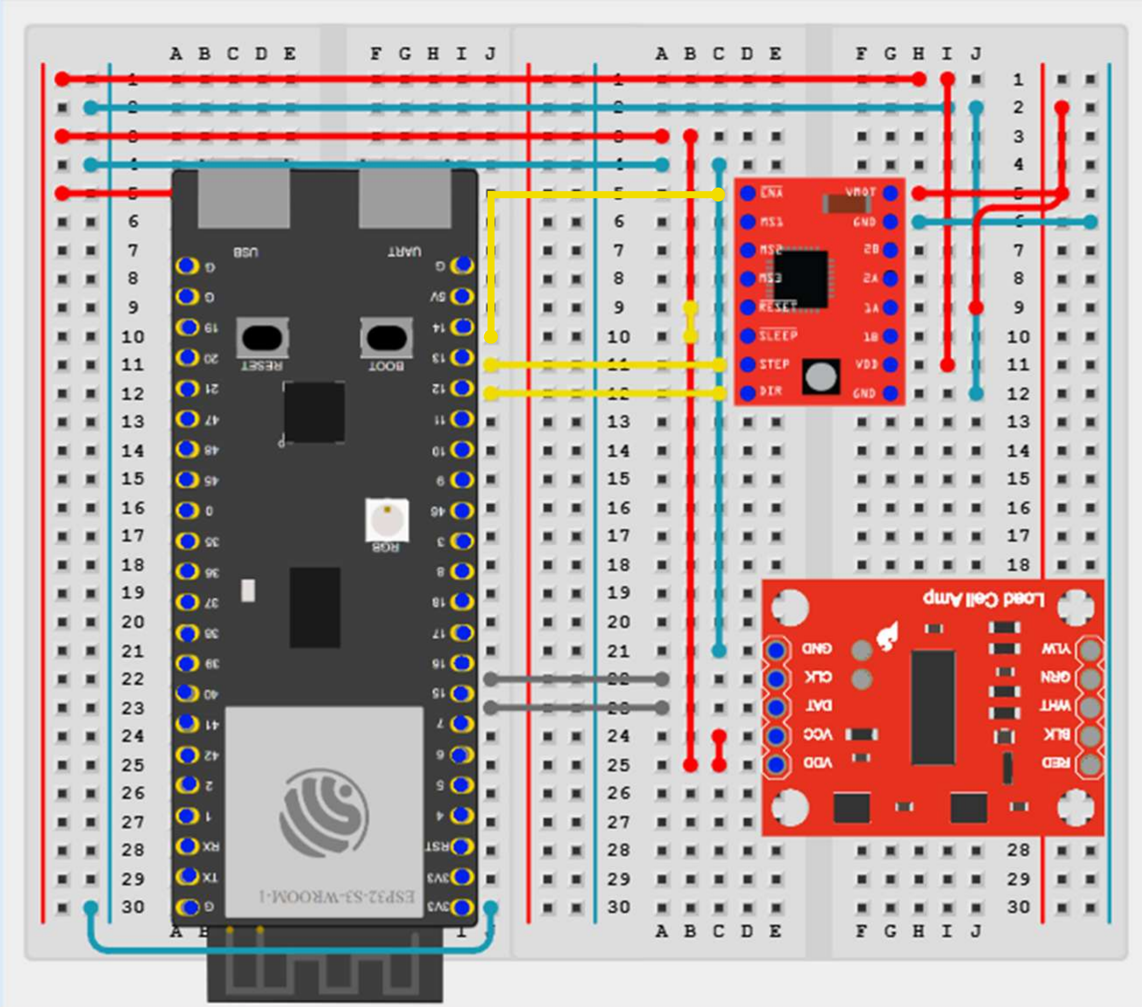
ESP32

3.

Place the ESP32, A4988 and HX711 in the shown positions (blue dots for connections)

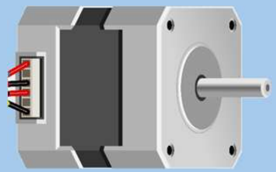
HX711 → ESP32:
DAT → GPIO 15
CLK → GPIO 16

A4988 → ESP32:
STEP → GPIO 13
DIR → GPIO 12



Stepper Motor and Load Cell

Breadboard Assembly



1x

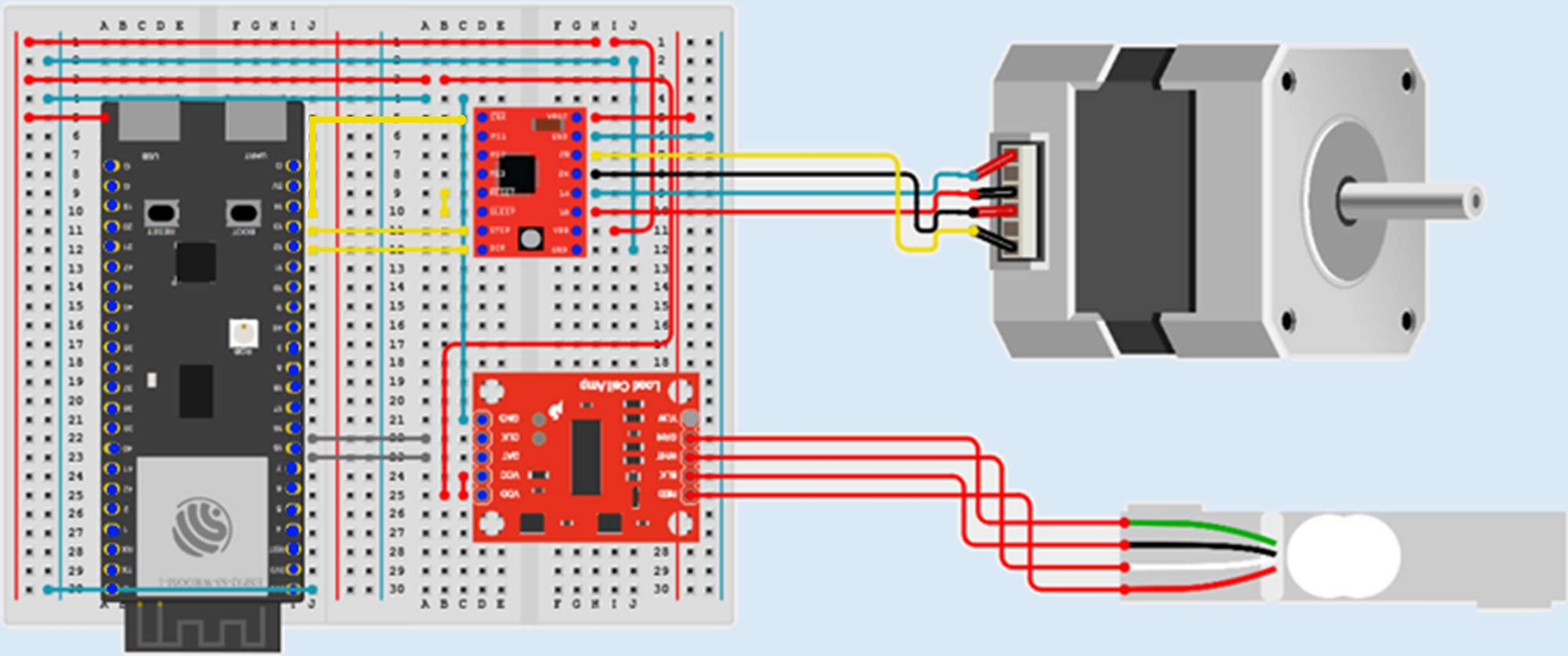
1x

Load Cell

Nema 17

4.

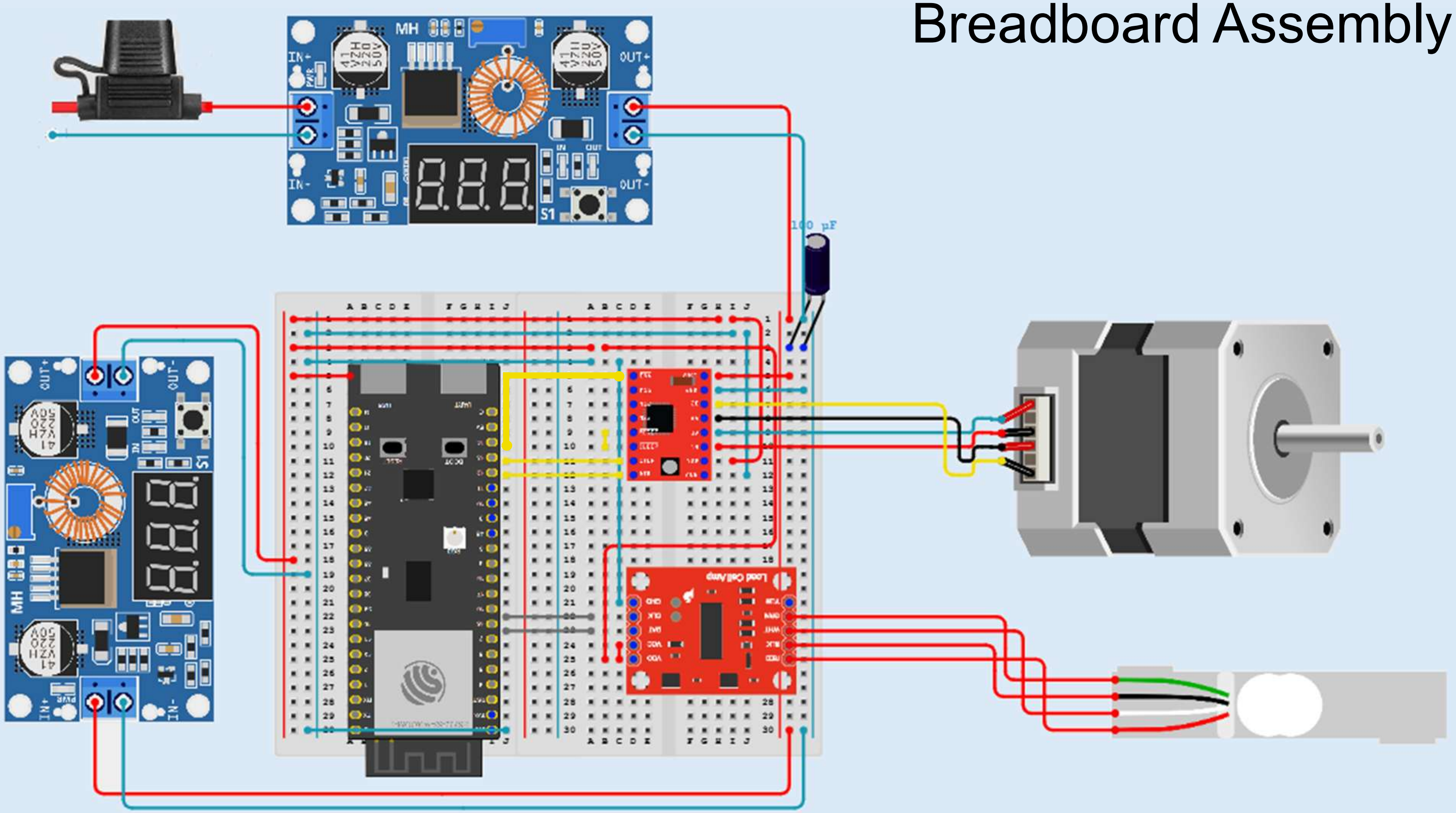
Connect the Nema 17, and the Load Cell to the breadboard as shown



Power provision

Breadboard Assembly

5. Connect the Buckboost units, the capacitor and the fuse as shown



2x

Buckboost

1x

100μF Decoupling Capacitor

1x

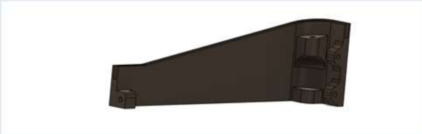
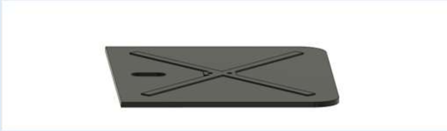
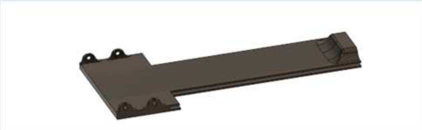
Fuse

6x

6. Set the value of the top Buckboost as shown to 12 V and the value of the left Buckboost to 3.3 V by turning the flathead screw on each Buckboost

Mechanical Assembly

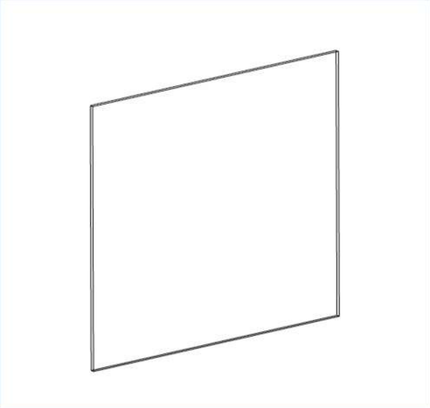
3D Printed Parts Catalogue

Part ID	Schematic	Part ID	Schematic	Part ID	Schematic
B Body 1		M Body 5		Rotary Wheel Cap	
B Body 2		M Body 6		T Body 1	
B Body 3		Pet Food Slant		T Body 2	
M Body 1		Rotary Wheel Holder		Hatch Closer	
M Body 2		Rotary Wheel		Hatch Knob	
M Body 3		Weight Scale Platform		Fuse Holder	
M Body 4		Pet Food Bowl			

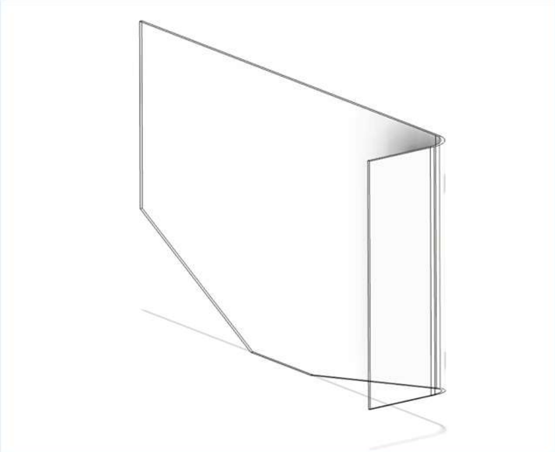
Mechanical Parts Catalogue

Part ID	Schematic	Part ID	Schematic
Ball Bearing 17mm x 30mm x 7mm		M4 x 10mm Bolt (8pcs)	
M3 x 6mm Bolt (8pcs)		M4 x 14mm Bolt (6pcs)	
M3 x 8mm Bolt (5pcs)		M4 x 20mm Bolt (12pcs)	
M3 x 16mm Bolt (2pcs)		M4 Nut (6pcs)	
M3 x 18mm Bolt (12pcs)		M5 x 18mm Bolt (2pcs)	
M3 x 20mm Bolt			
M3 Nut			

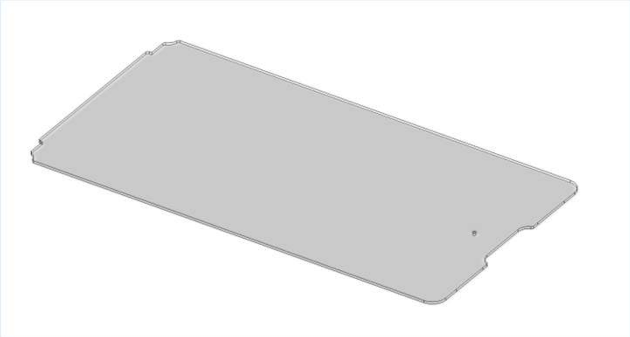
Acrylic Parts Catalogue



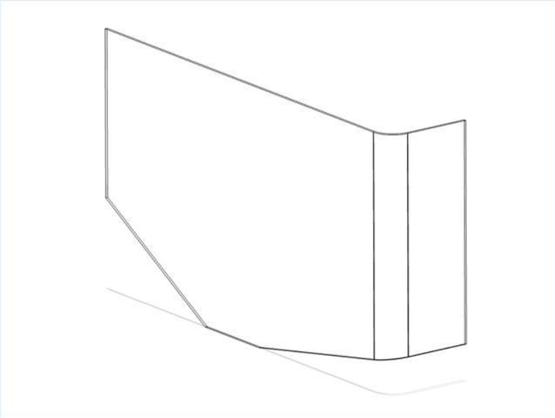
Front Acrylic



Left Acrylic



Hatch Acrylic

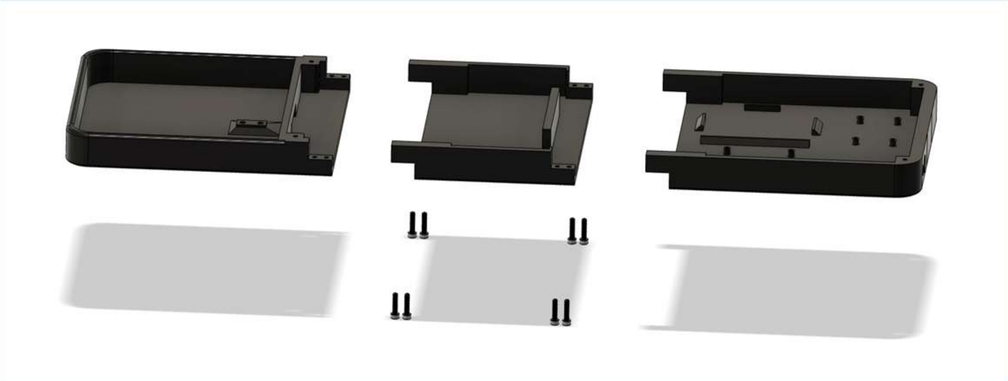


Right Acrylic

Base Assembly

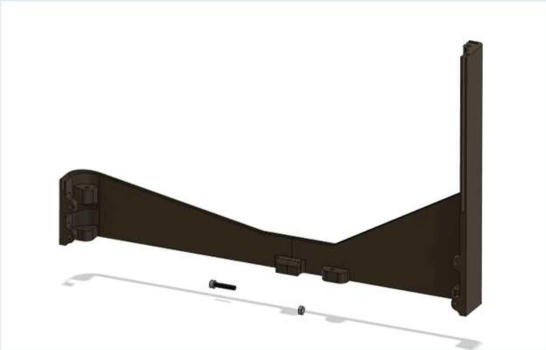
1.

Connect B Body 1, B Body 2, & B Body 3 with the use of 6 M3 x 18mm Bolts



2.

Connect M Body 1 & M Body 5 with the use of M4 x 20mm Bolt and M4 Nut.



Repeat **Step 2** with M body 2 & M Body 6

Body Assembly

3.

Connect M Body 1, 2, 3, 4, 5, & 6 with the use of 8 M4 x 10mm Bolts

4.

Connect Pet Food Slant, Rotary Wheel Holder, & 17mm x 30mm x 7mm Bearing, with the use of 2 M4 x 20mm

5.

Connect the Stepper Motor with the use of 4 M3 x 8mm Bolts

6.

Connect the Rotary Wheel to the stepper motor shaft and tighten from the side using M4 x 14mm

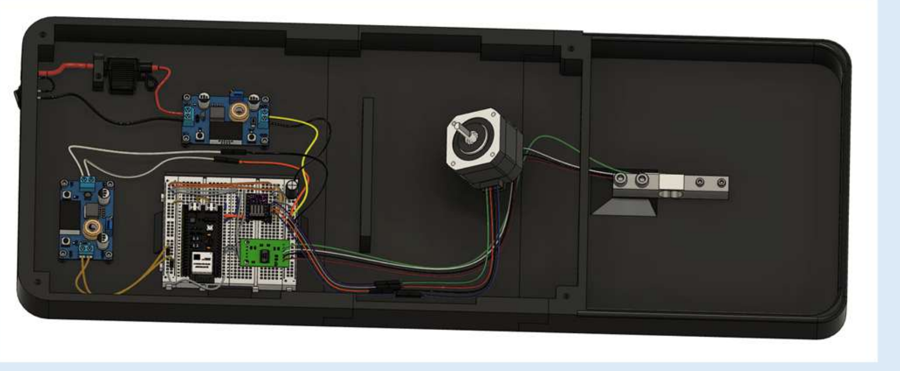
Body Assembly

7.

Connect M Bodies and Rotary Wheel Assemblies with the use of 6 M4 x 20mm Bolts

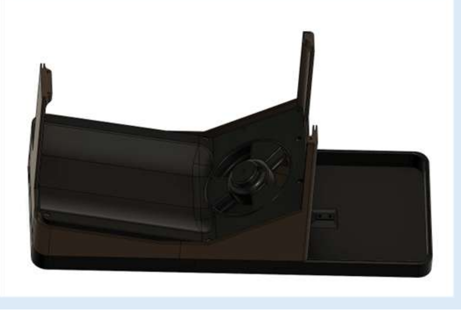
8.

Fit all the electronics accordingly



9.

Connect the B Bodies to the assembly with the use of 4 M3 x 18mm



10.

Install Acrylic Parts (Front, Left, Right)

Top Assembly

11.

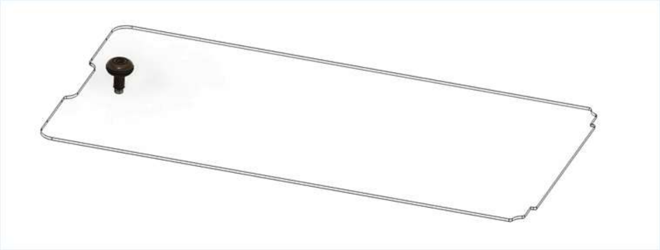
Connect T Body 1 & T Body 2 with the use of 2 M3 x 18mm and M3 Nuts

12.

Fit T Bodies on the main assembly and secure with the use of 4 M4 x 14mm Bolts

13.

Connect Hatch Acrylic to Hatch Knob with the use of M3 x 20mm and M3 Nut



14.

Install Acrylic Hatch and Hatch Closer with the use of M3 x 8mm Bolt

Bowl Assembly

15.

Connect Weight Scale Platform to the strain gauge with the use of 2 M3 x 16mm and M3 Nuts

16.

Place the Pet Food Bowl on the Weight Scale Platform

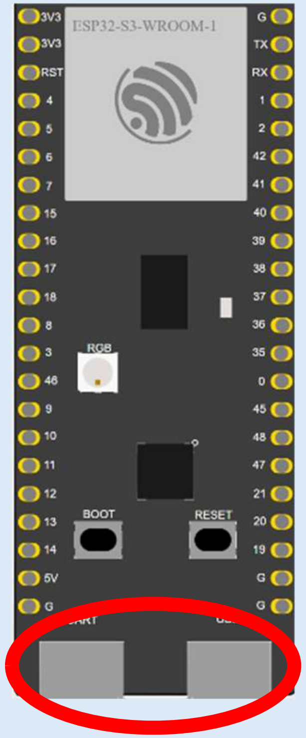
2. Firmware and App Installation

Software Installation

Recommended IDE: Visual Studio Code with Platform.IO
Project Files: https://github.com/amitchandran06/pet-feeder

ESP32 Firmware:

1. Install Platform.IO from this link: https://platformio.org/install, this is the development environment that will be used to flash the ESP32-S3
2. Pull/copy the code from the GitHub website above into a Platform.IO project
3. Plug into the ESP32-S3 port labelled "USB" (<u>not</u> COM or UART) and flash the code from the project onto the ESP32-S3



Connection Ports

Android App:

4. On an android mobile device, open the "app-release.apk" APK file in app-release and install the App for your device (currently no support for Apple Devices)